Question 3 Definite integral — 3 points

Evaluate the following integral and show the main steps.

$$\int_0^{\frac{\pi}{2}} x\cos x\,dx$$

Exact value _____

Question 4 Rolle's theorem — 6 points

Write your answers to all three parts on page 3.

(a) State Rolle's theorem with all hypotheses and its conclusion. Omit the proof. [2 pt]

(b) Give a function and an interval that satisfy every hypothesis. Verify the hypotheses and find a point satisfying the conclusion. [2 pt]

(c) Give a function and an interval for which exactly one hypothesis fails and the conclusion is false. Identify the failed hypothesis and justify your answer. [2 pt]

Sample Written Examination

Duration 2 hours • Maximum score 32 points

Surname and first name	Student ID
Signature	Score _____ / 32 (examiner use)

Answer all four questions. Questions 3 and 4 are on page 4. Record the requested results in the boxes. Draw the graph on page 2 and answer Question 4 on page 3. Show your working on the answer sheets and hand them in with this booklet.

Question 1 Parametric linear system — 8 points

Consider the following system in the unknowns x, y, z with real parameter k.

$$\begin{cases} x+y+z=1 \\ x+ky+z=2k-1 \\ x+y+(k+2)z=k \end{cases}$$

(a) Classify the system by the number of solutions as k varies. [4 pt]

(b) Find the solution set in every case where the system is consistent. [4 pt]

Unique solution	Infinitely many solutions	No solution
k ∈	k ∈	k ∈

Question 2 Complete study of a function — 15 points

$$f(x)=\begin{cases} x+1+\dfrac{1}{x+1} & x<0, \\ (2x-1)e^{-x} & x\geq 0. \end{cases}$$

(a) Find the domain and intercepts. Determine the sign of the function. [3 pt]

(b) Use limits to classify the discontinuities and find all asymptote equations. [4 pt]

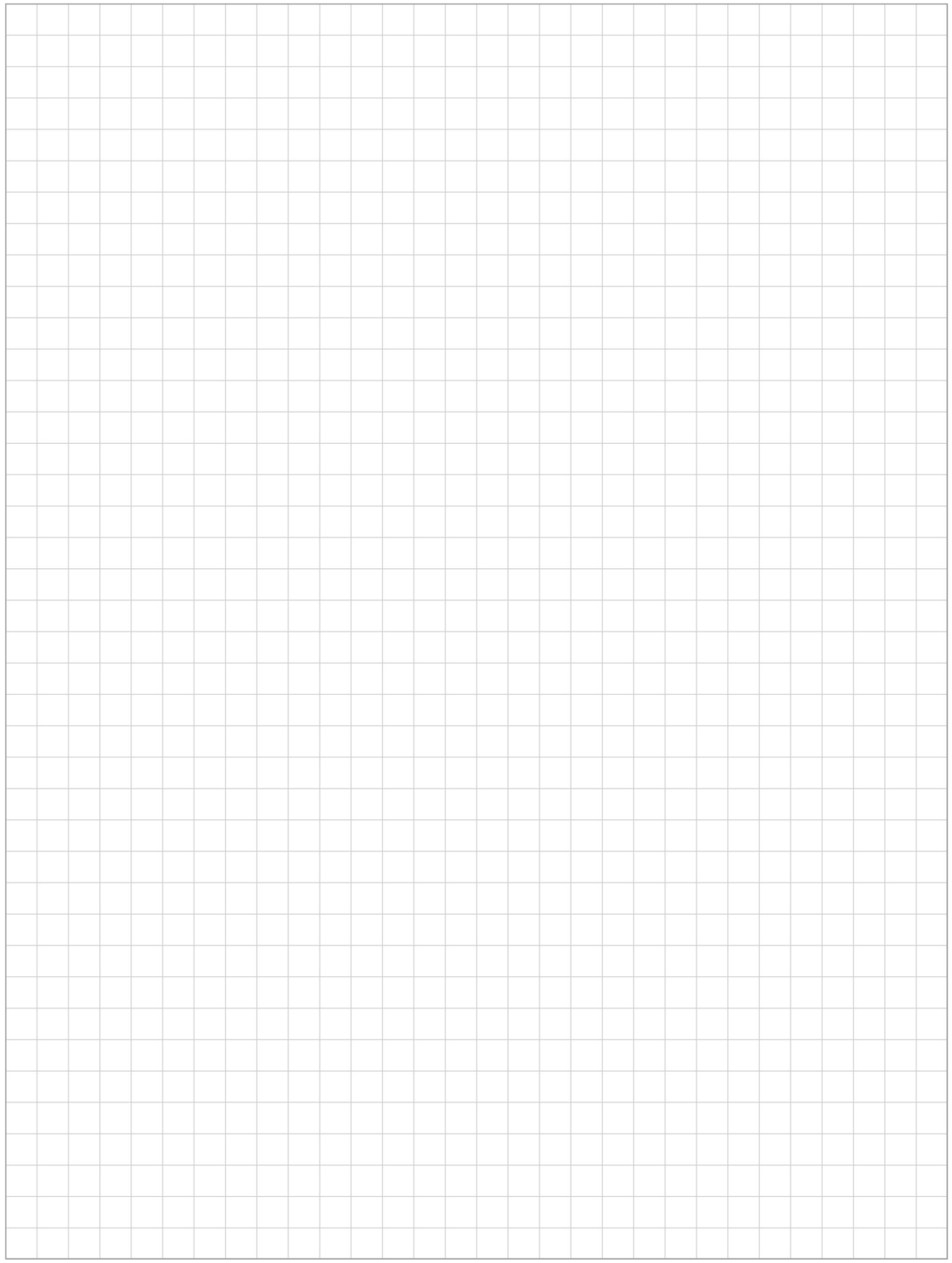
(c) Determine the intervals of increase and decrease and find any local extrema. [3 pt]

(d) Determine where f is concave up or down and find any inflection points. [3 pt]

(e) Sketch the graph on page 2. Mark the key points and all asymptotes. [2 pt]

Question 2 Graph of the function

Draw and label the axes and choose suitable scales. Mark intercepts, extrema, inflection points and any open or closed endpoints at discontinuities where applicable. Draw asymptotes as dashed lines.



Question 4 Written answers

Use the spaces below for each part of the question on page 4.

(a) Statement, hypotheses and conclusion

(b) Example and verification

(c) Counterexample and justification